The first column is a three-number code for the shape of Gargantua's vibrations and the picture is a still from a movie Romilly took, in my extrapolation of *Interstellar*, which verified that the vibrations had the predicted shape. The second column of data is the vibration frequency and the third is the rate at which this vibration dies out, as predicted by Teukolsky's equations.[32] The fourth and fifth columns show the difference between Romilly's observations and the theoretical predictions.

In my extrapolation Romilly finds a few anomalies, severe disagreements between his observations and the theory. He prints the disagreements in red. On page one of the data set (Figure 18.1), there is just one anomaly, but the disagreement is severe: thirty-nine times larger than the uncertainty in his measurements!

These anomalies might be helpful in "solving gravity" (learning how to harness the anomalies), Romilly thinks, in my extrapolation. He wishes he could transmit what he has learned to Professor Brand back on Earth, but the outbound communication link has been severed, so he's frustrated.

Even more, he wishes he could see inside Gargantua, to extract the crucial quantum data embedded in its singularity (Chapter 26). But he can't.

And he doesn't know whether the anomalies he observed are encoding some of the quantum data or not. Perhaps, with the hole spinning so rapidly, some of the quantum data leaked out through the horizon and produced the anomalies. Maybe Professor Brand could figure that out, if only Romilly could transmit the data to him.

I say a lot more later (Chapters 24–26) about gravitational anomalies, and quantum data from inside Gargantua as the key to harnessing the anomalies. But that's later. For now, let's continue our exploration of Gargantua's environs, turning next to Mann's planet.